

# Maré — morphology brief

Draft for PI review. Prepared to support proposal teams planning research in Maré.

## Purpose

MorphoFavela has already computed a detailed morphometric characterisation of Maré — built form, sky access, solar exposure, and geometry-derived ventilation tendencies — at 10 m grid and street-segment resolution across 3.36 km<sup>2</sup> of Maré's residential fabric. This brief summarises what exists so a proposal team can scope new work against it rather than starting from zero: what is measured, at what resolution, and what the PI can share. It is not a results paper — it is an inventory, written for researchers deciding whether and how to build on this morphological base.

*Study area: the IPP Territórios Sociais outline of Complexo da Maré (Prefeitura do Rio de Janeiro / IPP) — a single contiguous polygon covering the 16 communities recognised by Redes da Maré (boundary polygons: Prefeitura do Rio de Janeiro SABREN favela limits (2022) and conjuntos habitacionais, 5 of 16 name matches inferred rather than exact) plus the ground between them — streets, canals, open land inside the complex, not attributed to any one community. Marcílio Dias lies outside the outline and is excluded from every figure and count below; the remaining 15 communities, plus the ground between them, make up the 3.36 km<sup>2</sup> reported throughout.*

## Maré at a glance

|                                               |                          |
|-----------------------------------------------|--------------------------|
| Area                                          | 3.36 km <sup>2</sup>     |
| Typology                                      | flatland                 |
| Buildings (site + context)                    | 39,322                   |
| 10 m grid cells                               | 33,488 (74.1% built)     |
| Mean building height (grid mean, built cells) | 7.1 m                    |
| Annual mean street-level sun                  | 6.61 h/day               |
| Dominant wind sector                          | SE (26% of observations) |

Maré is the largest of MorphoFavela's five campaign sites and the flattest — a flatland settlement built out over former mangrove and reclaimed land, organised around several distinct planned-housing projects and later infill. Its scale and its structural regularity (large, near-saturated block interiors) make it a useful counterpoint to the hillside sites in the same campaign.

## Data inventory

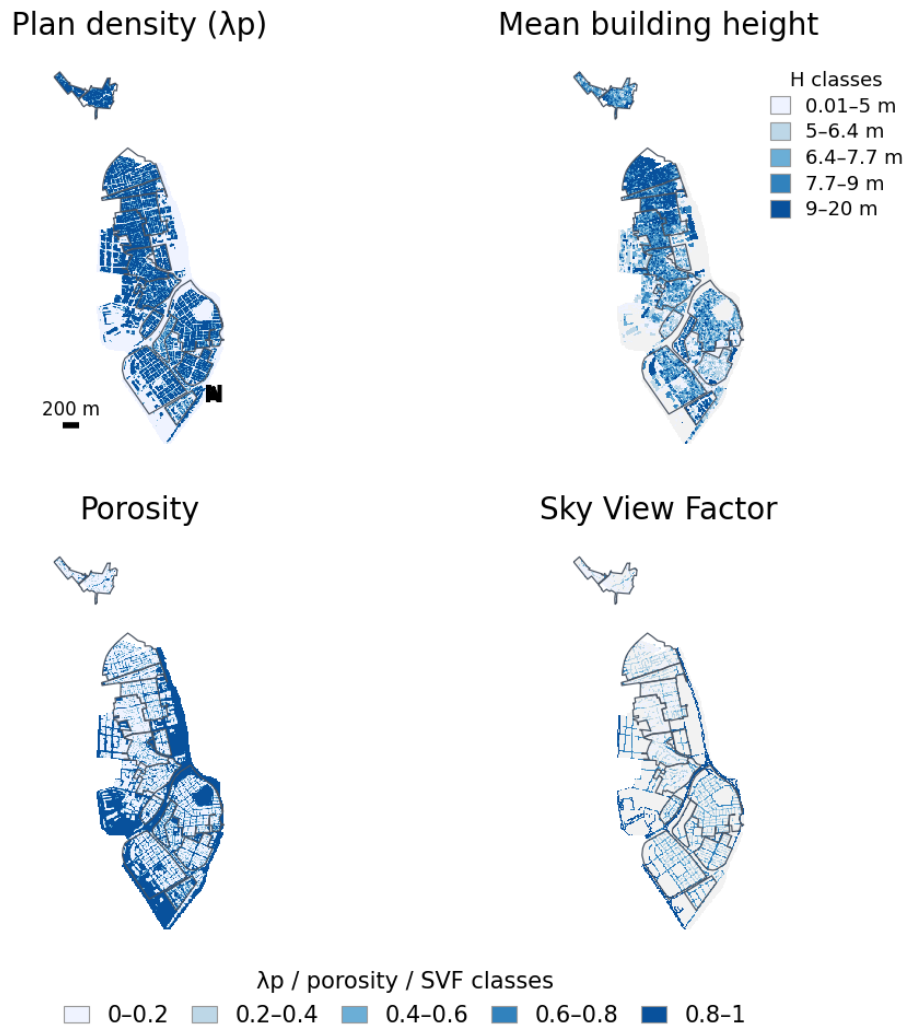
| Layer                                   | Resolution / unit                       | Provenance                                                                         | Derived or restricted | Shareable form                                       |
|-----------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------|
| Building footprints (input)             | per-building polygon                    | municipal cadastre (2019) + LiDAR height attributes                                | restricted input      | not shareable; pipeline is open                      |
| Digital terrain model (input)           | 5 m raster                              | municipal DTM                                                                      | restricted input      | not shareable; pipeline is open                      |
| Morphometric grid                       | 10 m cell, 33,488 cells                 | derived ( $\lambda_p$ , height, porosity, SVF per cell; terrain columns held back) | derived aggregate     | shareable, de-georeferenced, without terrain columns |
| Street-level SVF and solar hours        | per street sample point (71,247 points) | derived (ray-cast SVF + sun-position accumulation)                                 | derived aggregate     | shareable, de-georeferenced                          |
| Street-segment SVF                      | per segment (1,917 segments)            | derived, aggregated from street points                                             | derived aggregate     | shareable, de-georeferenced                          |
| Geometry-derived ventilation tendencies | per built cell                          | derived ( $\lambda_f$ regime, lateral depth, wind exposure)                        | derived aggregate     | shareable, de-georeferenced                          |
| Wind climatology                        | eight-sector rose                       | Galeão (SBGL) METAR via Iowa ASOS, 89,439 observations, 2015–2024                  | derived aggregate     | shareable                                            |

Per-cell georeferenced layers (coordinates attached) are available on request under the project's data-sharing conditions; see Availability and terms, below.

## Built form

Four indicators characterise the built fabric at 10 m resolution: plan density ( $\lambda_p$  — the fraction of each cell covered by building footprint), mean building height, porosity (the complement of  $\lambda_p$ ), and Sky View Factor (SVF, the fraction of hemispheric sky visible at pedestrian height).

### Maré — built-form indicators (10 m grid)



**Figure 1.** Plan density, mean height, porosity, and Sky View Factor across Maré's grid, each band-classified in five discrete classes with class edges in the legend. No basemap; scale bar and north arrow only.

Across all 33,488 grid cells (built and unbuilt), median  $\lambda_p$  is 0.577 (IQR 0–1) and median SVF is 0.451 (IQR 0.244–0.709). Restricted to the 24,820 built cells (74.1% of the grid), median mean height is 7.04 m (IQR 5.38–8.65 m) and median height variability ( $\sigma H$ ) is 1.93 m (IQR 1.34–2.53 m) — consistent with the predominantly two- to three-storey construction typical of the campaign. Median porosity on built cells is 0.18 (IQR 0–0.571).

The high all-cell  $\lambda_p$  median reflects the study area itself, not the whole bairro: restricted to the IPP Territórios Sociais outline (Redes da Maré's 15 communities plus the ground between them), most grid cells fall on or immediately next to a building, whereas the bairro's open land and low-density fringes outside the outline are excluded from the figure entirely rather than folded in and pulling it down. The low built-cell porosity is consistent with this: large, near-saturated housing-project block interiors dominate the study area's built fabric.

## Maré — indicator distributions

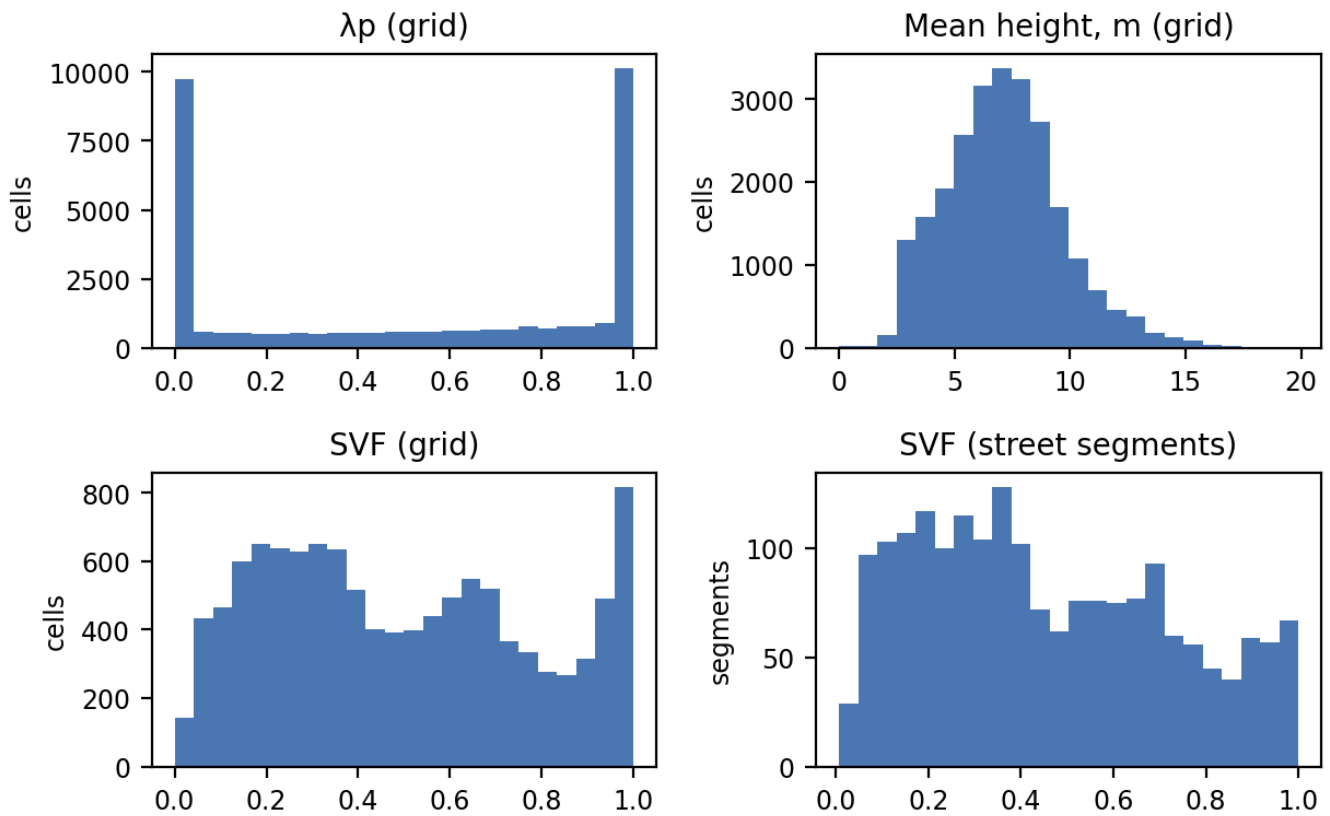


Figure 2. Distributions of  $\lambda_p$ , mean height, SVF (grid), and street-level SVF (segments) across Maré, as histograms.

## Sky access and sun

Street-level Sky View Factor and direct-sun hours are computed at 71,247 passageway sample points (aggregated to 1,917 street segments) via ray-casting and sun-position accumulation on four reference dates. Median street-segment SVF is 0.4.

### Maré — street-segment Sky View Factor

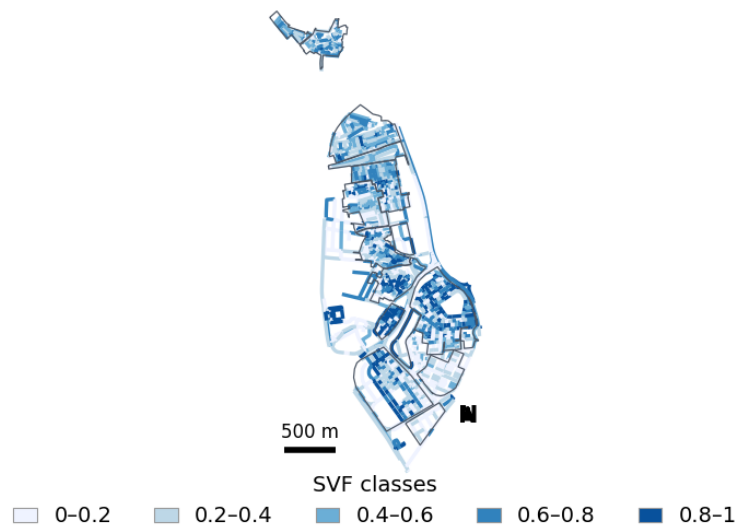


Figure 3. Sky View Factor by street segment, five discrete classes with class edges in the legend. No basemap, no coordinates.

Mean street-level direct-sun hours are 4.7 h/day at the winter reference date, 6.61 h/day on the unweighted annual proxy (mean of the four reference dates), and 8.27 h/day in summer — a seasonal range of 3.57 h. Against the 2 h/day

daylight-adequacy floor of the Athens Charter (1943), Point 26, 31.1% of street points fall below the floor at the winter reference date, 11% on the annual proxy, and 5.94% in summer.

## Geometry-derived ventilation potential

Three geometric tendencies — never air-exchange adequacies — extend the built-form picture with two more degrees of ventilation-relevant structure: vertical enclosure ( $\lambda f$  regime), lateral depth into contiguous fabric, and directional wind exposure. Across Maré's 24,820 built cells, 67.7% meet the campaign's high-enclosure threshold on frontal-area density  $\lambda f$  (vertical constraint), 53.2% sit at or beyond the campaign's median lateral open-edge distance (lateral constraint), and 75.5% have a directional wind-exposure ratio at or above the isotropic baseline (directional constraint). Counting how many of these three independent axes each cell trips: 8.45% trip none, 22% trip one, 34.3% trip two, and 35.3% trip all three.

The wind climatology behind the directional axis comes from Galeão (SBGL) METAR via Iowa ASOS (89,439 observations, 2015–2024, calm fraction 3.69%). The dominant sector is SE at 26% of observations; the median directional wind-exposure ratio across Maré's built cells sits near the isotropic baseline, at 1.01.

### Maré — wind-direction frequency (Galeão (SBGL) METAR via Iowa ASOS)

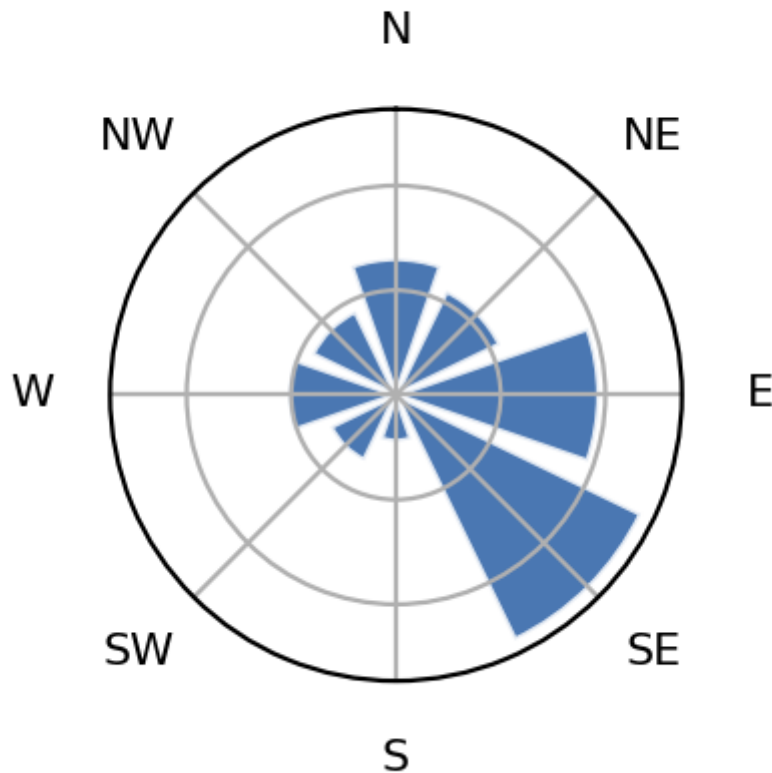


Figure 4. Measured wind-direction frequency at Galeão (SBGL) METAR via Iowa ASOS.

A morphometric roughness-length estimate is also available per patch, but its published method envelope is wide at Maré's density: across 23 sampled patches, 60.9% fall outside the calibration range of the published drag-partition methods used, and 17.4% hit the estimator's lower floor. No absolute roughness value is reported here for that reason.

These geometric tendencies are a pre-simulation prioritisation surface, not a ventilation verdict. A wind-simulation study of two Maré patches is a parked companion track with no results to report at this time.

## Maré's communities

Redes da Maré recognises 15 communities within the study area; MorphoFavela's grid and the study-area sun/sky run (wp04\_mare\_studyarea\_20260924T201015Z) can both be labelled by which community each cell or sample point falls in (src.sites.territory.label\_subunits), including the streets, canals, and open ground between communities that the study area counts but no single community claims ("between communities"). The panel and table below break the

site-wide figures above down to that resolution — density, height, footprint, sky view, winter sun, and annual irradiation, one row per community, ordered north to south by each community's mean grid-cell position. This is a geographic ordering, not a ranking: no row is privileged over another, and the two housing-project communities and the older, denser communities are described side by side rather than contrasted as better or worse.

### Maré — per-community morphology (north → south, not ranked)

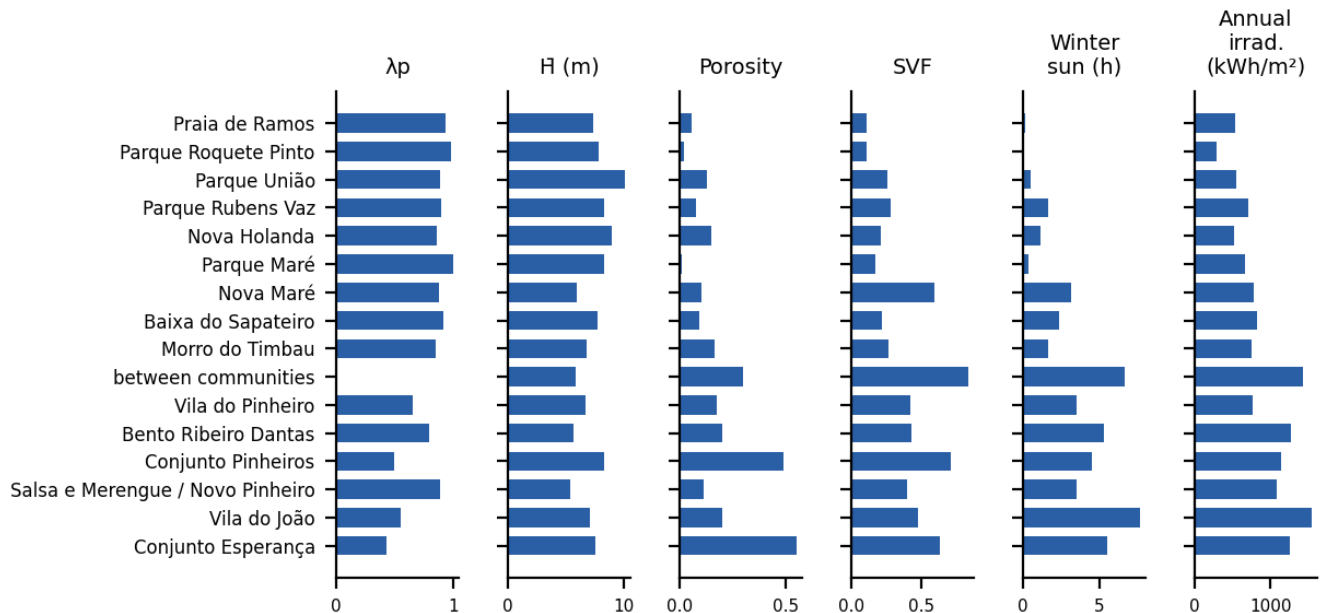


Figure 5. Per-community morphology, one bar-chart column per indicator, rows ordered north to south.

| Community                        | Cells  | $\lambda_p$ | FAR   | $\bar{H}$ (m) | $\sigma H$ (m) | Porosity | SVF   | Winter sun (h) | Annual irradi. (kWh/m²) |
|----------------------------------|--------|-------------|-------|---------------|----------------|----------|-------|----------------|-------------------------|
| Praia de Ramos                   | 399    | 0.935       | 1.81  | 7.32          | 2.04           | 0.0577   | 0.104 | 0.167          | 535                     |
| Parque Roquete Pinto             | 851    | 0.985       | 2.08  | 7.78          | 2.03           | 0.0177   | 0.11  | 0              | 288                     |
| Parque União                     | 1,958  | 0.892       | 2.39  | 10            | 1.98           | 0.126    | 0.255 | 0.5            | 553                     |
| Parque Rubens Vaz                | 747    | 0.898       | 1.99  | 8.24          | 2.15           | 0.0746   | 0.278 | 1.67           | 703                     |
| Nova Holanda                     | 1,502  | 0.86        | 1.94  | 8.93          | 1.97           | 0.148    | 0.209 | 1.17           | 525                     |
| Parque Maré                      | 1,406  | 1           | 2.13  | 8.27          | 2              | 0.0117   | 0.173 | 0.333          | 667                     |
| Nova Maré                        | 475    | 0.88        | 1.35  | 5.95          | 2.21           | 0.103    | 0.594 | 3.17           | 782                     |
| Baixa do Sapateiro               | 1,242  | 0.914       | 1.81  | 7.69          | 2.08           | 0.0946   | 0.215 | 2.33           | 821                     |
| Morro do Timbau                  | 1,249  | 0.855       | 1.52  | 6.79          | 1.99           | 0.166    | 0.267 | 1.67           | 757                     |
| between communities              | 13,467 | 0           | 0.992 | 5.85          | 1.61           | 0.301    | 0.83  | 6.67           | 1430.0                  |
| Vila do Pinheiro                 | 3,694  | 0.656       | 1.39  | 6.65          | 2.01           | 0.173    | 0.419 | 3.5            | 772                     |
| Bento Ribeiro Dantas             | 506    | 0.799       | 1.08  | 5.63          | 1.88           | 0.202    | 0.431 | 5.33           | 1280.0                  |
| Conjunto Pinheiros               | 601    | 0.5         | 0.884 | 8.29          | 5.04           | 0.487    | 0.706 | 4.5            | 1140.0                  |
| Salsa e Merengue / Novo Pinheiro | 1,218  | 0.889       | 1.18  | 5.4           | 1.61           | 0.113    | 0.392 | 3.5            | 1080.0                  |
| Vila do João                     | 3,485  | 0.556       | 1.42  | 7.05          | 2.18           | 0.199    | 0.472 | 7.67           | 1540.0                  |
| Conjunto Esperança               | 688    | 0.425       | 0.833 | 7.53          | 3.79           | 0.55     | 0.627 | 5.5            | 1260.0                  |

### A Maré-internal fabric grouping

"Maré among the five campaign sites", below, describes the campaign's fabric-vector clustering, fit once across all five sites and applied here — one cluster accounts for 75.6% of Maré's built cells, because that fit has to separate Maré's fabric from four other settlements' fabric at the same time, not because Maré's own fabric is undifferentiated. Refitting the same clustering method restricted to Maré's own 29,195 built cells only — never pooled with the other four sites, and with  $k$  chosen freshly rather than inherited — gives a different picture:

| Group          | Share of built cells |
|----------------|----------------------|
| Fabric group 1 | 4.5%                 |
| Fabric group 2 | 19.4%                |
| Fabric group 3 | 9.29%                |
| Fabric group 4 | 37.3%                |
| Fabric group 5 | 29.5%                |

BIC does not settle to an interior minimum over the tested range ( $k = 2-8$ , matching the campaign fit's own range) – it decreases monotonically, so a plain minimum would just report the range's upper bound.  $k$  is instead chosen by the geometric elbow of the BIC curve.

The largest Maré-internal group holds 37.3% of built cells, well below the campaign fit's 75.6%, which is expected: a fit run only on Maré is free to resolve distinctions within Maré's fabric that a five-site fit spends its clusters separating sites instead. Both fits are legitimate for what they each answer – the campaign fit for "where does Maré's fabric sit among the five sites", this one for "how much internal variation does Maré's own fabric have" – and neither supersedes the other.

## Maré among the five campaign sites

MorphoFavela's five-site campaign spans hillside and flatland informal settlements in Rio de Janeiro. Maré is one of two flatland sites in the campaign, distinguished among them by its scale and by the near-uniform block interiors of its planned-housing sections. This section positions Maré within that typology; it is not a ranking of the five sites. Unlike the rest of this brief, the composition below covers Maré's whole analysed data extent (the bairro), not the study area (the IPP Territórios Sociais outline) – the underlying fabric-cluster fit is shared across all five campaign sites and is not re-run per community.

The campaign's fabric-vector clustering assigns each built cell to one of six recurring fabric clusters. Maré's composition:

| Fabric cluster | Share of built cells |
|----------------|----------------------|
| A              | 7.54%                |
| B              | 3.65%                |
| C              | 1.16%                |
| D              | 0.298%               |
| E              | 11.8%                |
| F              | 75.6%                |

Maré's fabric is dominated by one cluster with plan density near its maximum, associated with the tight, near-fully-covered block interiors of the original housing-project layout; a second cluster, common across all five campaign sites, is present as a secondary component. Two of the six clusters occur only where flat buildable land exists, which is why they concentrate at the two flatland sites rather than recurring campaign-wide.

## Availability and terms

Derived aggregates – the morphometric grid, street-level SVF and sun-hour summaries, geometry-derived ventilation tendencies, and the wind climatology – can be shared, including a de-georeferenced version of the grid (relative cell indices, no coordinates). The underlying building cadastre and LiDAR-derived terrain inputs cannot be shared in any form that reconstructs them (licence-restricted third-party data). Per-cell georeferenced layers are available on request under the project's conditions.

Contact: `{pi_contact}`

## Methods notes and references

- Sky View Factor: Tregenza sky discretisation, ray-cast at pedestrian height from passageway sample points.
- Frontal-area density and enclosure-threshold classification: Oke (1988); Stewart & Oke (2012), local climate zones.
- Daylight-adequacy floor: Athens Charter (1943), Point 26.
- Roughness-length estimation: morphometric drag-partition methods, method envelope only; not independently validated at time of writing.